# CO1 AT3

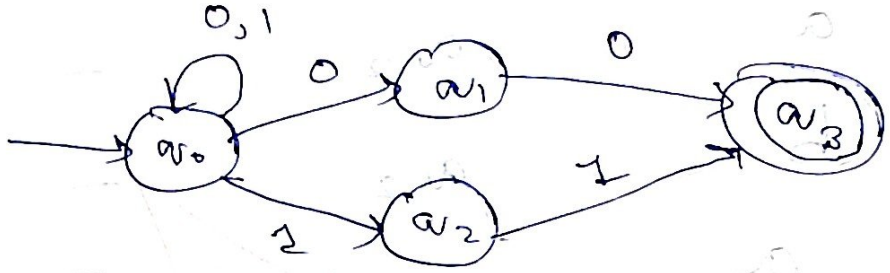
2.

Given

$\Sigma = \{0, 1\}$

last digits 00 or 11

three state $q_0, q_1, q_2, q_3$




Formal definitions

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

To check

0 1 0 1 1 is accepted by NFA

The path is

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_3$

Hence it is

0 1 0 1 1 is accepted

21. 001 as Substring

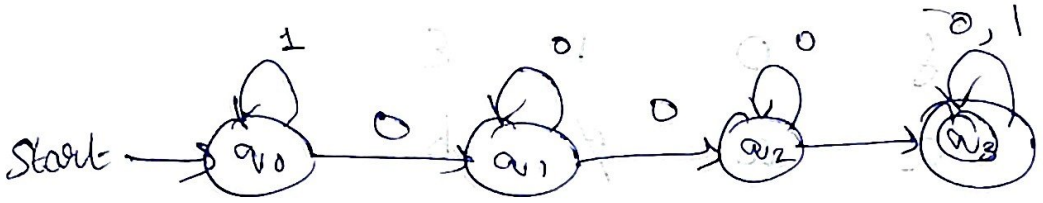
$$\Sigma = \{0, 1\}$$

formal definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$




001 as a Substring
to check it using an
example: 1001

(b)

$q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3$

1001 is
accepted.